

Solution Manual

Steven, Derek, Andy

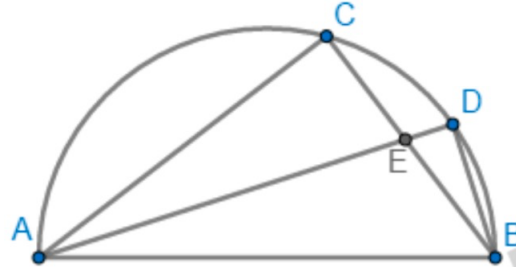
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Steven

Solution Q1 Geometry

Consider semicircle with diameter AB , C is a point, D is the midpoint of arc BC . AD meets BC at E . If $CE = 3$, and $BD = 2\sqrt{5}$, find AB .

Proposed by Steven Xia.



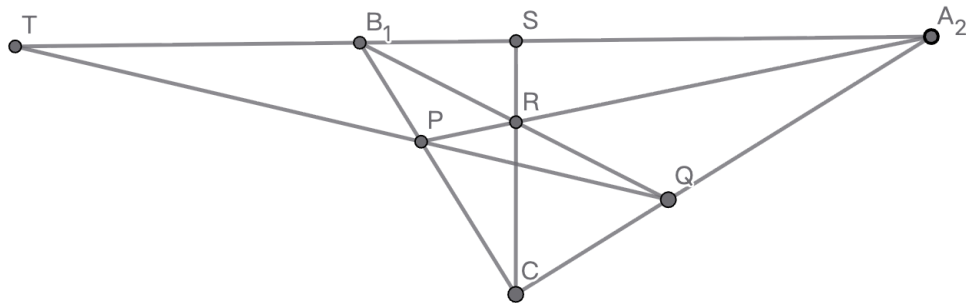
Answer: 10

Solution. Extend AC and BD to meet at P , $DP = BD = 2\sqrt{5}$. $CEDP$ con-cyclic. Consider power of point at B , we get $BE \cdot BC = BD \cdot BP = 40 \implies BE = 5, CP = 4, AB = 10$.

Solution Q2 Geometry

Consider right $\triangle ABC$ with $\angle C = 90^\circ$, $AB = 10$, and $AC = 8$. $CP = CQ = 2$. BQ and AP meet at R . CR meet AB at S . QP meet AB extension at T . Find TS .

Proposed by Steven Xia.



Answer: 24

Solution. Notice A, S, B, T is a harmonic bundle, so $\frac{AS}{BS} = \frac{AT}{BT}$. (Or alternatively Ceva and Menelaus with respect to $\triangle ABC$.) From Ceva in $\triangle ABC$, we get $\frac{AS}{BS} = \frac{3}{2} \implies \frac{10+BT}{BT} = \frac{3}{2} \implies BT = 20, ST = 4 + 20 = 24$.

Derek

Q3 Combinatorics Ryan has three six-sided dice, one fair, one that rolls threes $\frac{6}{7}$ of the time, and each of the other five sides equally with probability $\frac{1}{35}$ of the time, and one that rolls threes $\frac{2}{7}$ of the time, and each of the other five sides equally with probability $\frac{1}{7}$ of the time. He selects one of the dice at random. The probability he rolls a three given the first two rolls are threes is $\frac{p}{q}$. What is the remainder when $p + q$ is divided by 1000?

Proposed by Derek Hu.

Answer: 895

Solution. Using conditional probability, we find that the probability is

$$\frac{\frac{1}{3}(\frac{1}{6})^3 + \frac{1}{3}(\frac{6}{7})^3 + \frac{1}{3}(\frac{2}{7})^3}{\frac{1}{3}(\frac{1}{6})^2 + \frac{1}{3}(\frac{6}{7})^2 + \frac{1}{3}(\frac{2}{7})^2} = \frac{6961}{8934}$$

$6961 + 8934 = 15895$, thus the answer is 895. I apologize for using such atrocious numbers.

Q4 Number Theory

For all integers $1 < n < 500$, there exists a number that is divisible by all positive integers less than or equal to n except m and $m + 1$. Find the sum of all possible m .

Proposed by Derek Hu.

Answer: 447

Solution. We need m and $m + 1$ to be both in the form p^k , otherwise some factors of m or $m + 1$ also won't divide into n . Since m and $m + 1$ are of different parity, we know that one of them must be in the form of 2^k . Then, we can bash out the m that work: 1, 2, 3, 4, 7, 16, 31, 127, 256. Adding these up yields 447.

Andy

Solution Q5 Number Theory

Let a and b be positive integers satisfying

$$\text{lcm}(a, b) - \text{gcd}(a, b) = 23.$$

Find the sum of all possible values of $a + b$.

Proposed by Andy.

Answer: 129

Solution. Let $d = \gcd(a, b)$ and set $x = \frac{a}{d}$, $y = \frac{b}{d}$. It is well known that $\text{lcm}(a, b) = dxy$. Thus,

$$d(xy - 1) = 23,$$

so either $d = 1$ or $d = 23$.

- If $d = 1$, then $xy = 24$, so $(a, b) = (1, 24), (2, 12), (3, 8), (4, 6)$ or reverses, giving $a + b = 25, 14, 11, 10$. - If $d = 23$, then $xy = 2$, so $(a, b) = (46, 23)$ or reverse, giving $a + b = 69$.

Adding up the values gives $10 + 11 + 14 + 25 + 69 = 129$

Solution Q6 Number Theory

Let n be a positive integer. Find the sum of all possible values of

$$\gcd(5n + 11, n + 7).$$

Proposed by Andy.

Answer: 60

Solution. By the Euclidean Algorithm,

$$\gcd(5n + 11, n + 7) = \gcd(n + 7, 24).$$

Thus, all possible values are the divisors of 24, namely 1, 2, 3, 4, 6, 8, 12, 24. Then, adding them up gives $1 + 2 + 3 + 4 + 6 + 8 + 12 + 24 = 60$

Bonus Problem. Solve this problem for arbitrary $\gcd(an + b, cn + d)$.